Recipe 6

Large Scale Variation

Motto

The reference is the illusion

Your genome is real

The reference genome is not "reality"

It is an *artificial* construct we use to describe reality.

We need the *reference genome* to be able to *refer* to the same thing.

🤔 When does it make sense to compare two *real* genomes via a *third* artificial genome?

Are changes *transitive*: `A vs B` and `A vs C` what do we know about `B vs C`?

Information is carried by reads or read pairs that pass across a junction

Assume that a real genome has an insertion of Cs into a genome

```
REALITY ->       ...TTGCATGCTAGATCCCCCCCCCCGACATTTTTCACC...
```

The "junction" is in the reference!

```
REFERENCE ->     ...TTGCATGCTAGATGACATTTTTCACC...
INSERTION ->                     |
```

The reference is

a "funhouse" mirror

evertyhing is backward

Sequencing Simulations

Each recipe example will do the following:

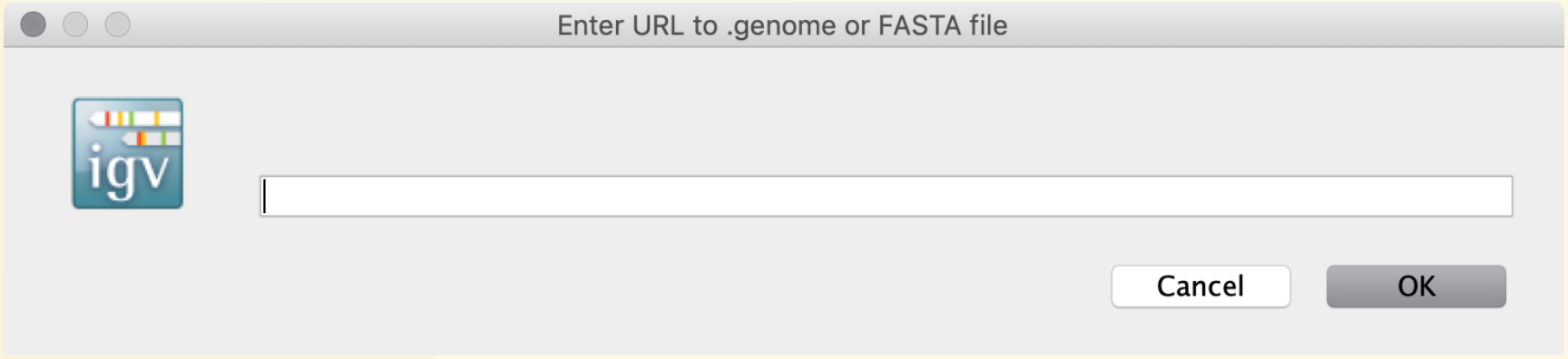
1. Modify a genome to be different from the reference
2. Simulates reads from genome
3. Align against simluated reads against the reference

Challenge: what can you determine?

1. Load the genome in IGV

The recipe allows you to load the genome via the URL.

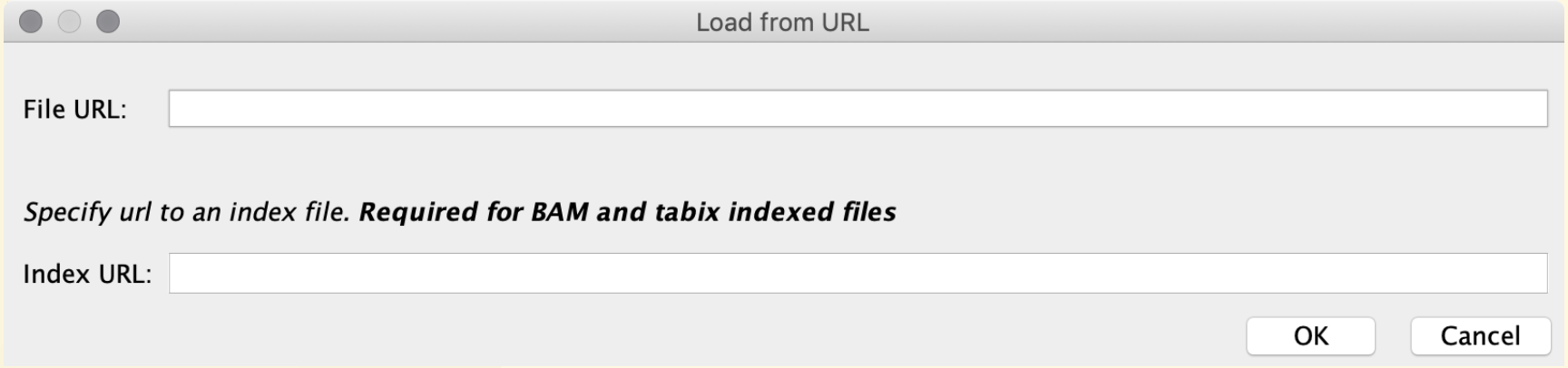
`Genomes -> Load Genome from URL`



2. Visualize BAM files

Copy the link to each alignment file and put it into

`File -> Load from URL`



The data will show up in IGV.

3. Investigate each BAM



Each file is different

We have made a different large scale modification in each bam file.

See if you can identify what it is visually.

Check the code for an explanation of what the change (if any) was.